Consent and Demographic Questions

Thank you for your willingness to participate in this study – we are truly grateful.

This project explores the ability of large language models (such as ChatGPT and Gemini) to interpret and respond to medical images and information. To fully assess these models, we would like to establish a baseline of how a clinical expert interprets and responds to the same task.

We will first ask a few general questions including whether you are willing to participate and about your clinical background. Next, we will show you a series of images and ask you to evaluate whether the stated clinical feature is visible and accurately describes what you observe in the image, as well as your confidence level with this task. We will end with a few questions about your experience evaluating images.

We anticipate that participation will take most people less than 20 minutes.

We encourage you to take this survey on your laptop/desktop as it may be easier to view images. However, you can still take it on your smart device.

Please do not use the internet or textbooks or look up answers. We are interested in your instinctive response to each question. Once you answer a question, you will not be able to go back.

No identifiers will be collected as part of the survey. All individual results will be maintained in an

anonymous format. You will not receive your individual results. We plan to describe/publish only deidentified results. Your participation is voluntary and will not be compensated. We have been granted exemption from IRB review by the National Institutes of Health (NIH). By continuing with the survey, you agree to take part. If you have questions, encounter any difficulties or change your mind about participation, please contact Rebekah Waikel at rebekah.waikel@nih.gov.

We will begin by asking you demographic questions.

Are you willing to participate in this study?

- ○ Yes
- ○ No

What is your primary practicing clinical specialty?

- ○ Medical Genetics
- ○ Molecular Genetics or Laboratory Genetics and Genomics
- ○ Other

If other, please enter your practicing clinical specialty.

What are your clinical credentials?

- [ ] Physician (MD/DO/equivalent)
- [ ] Genetic Counselor (MS/ScM/MSC)
- [ ] Physician Assistant (MPAS/equivalent)
- [ ] Nurse Practitioner (MSN or DNP)
- [ ] Board certified in Molecular Genetics or Laboratory Genetics and Genomics (PhD or MD/equivalent)
- [ ] Other

If other, please enter your clinical credentials?

Are you a board certified/eligible practicing physician or currently in training?

- ○ Board certified/eligible practicing clinician
- ○ Trainee: resident or fellow

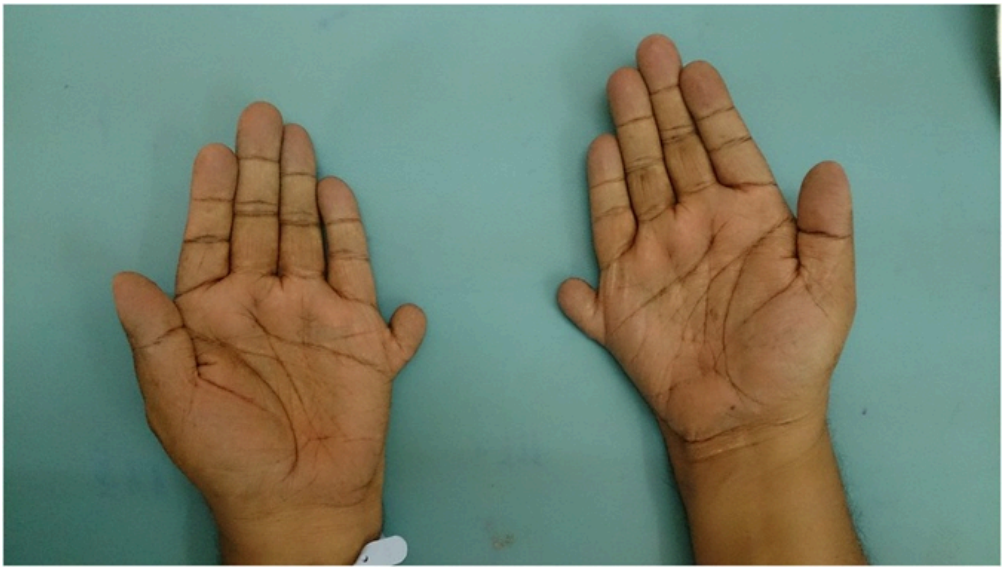
For how long (since you became board eligible/completed medical genetics training) have you been practicing?

- ○ Less than 1 year
- ○ 1 to 5 years
- ○ 5 to 10 years
- ○ More than 10 years

Genetic Conditions

For the following questions, critically evaluate each image and description provided, then select whether the stated human phenotypic abnormality is present in the image. You will then be asked to select your confidence level with your selected answer.

Set 1A



This medical image shows a patient with Bardet-Biedl syndrome. Please evaluate whether the following clinical feature is visible and accurately describes what you observe in the image -- Feature: postaxial polydactyly.

- ○ Yes
- ○ No

How confident are you with your answer?

- ○ Highly confident
- ○ Confident
- ○ Somewhat confident
- ○ Not confident



This medical image shows a patient with Split-Hand/foot malformation. Please evaluate whether the following clinical feature is visible and accurately describes what you observe in the image -- Feature: 2-3 toe syndactyly.

- ○ Yes
- ○ No

How confident are you with your answer?

- ○ Highly confident

- ○ Confident
- ○ Somewhat confident
- ○ Not confident



This medical image shows a patient with Achondroplasia. Please evaluate whether the following clinical feature is visible and accurately describes what you observe in the image -- Feature: arachnodactyly.

- ○ Yes
- ○ No

How confident are you with your answer?

- ○ Highly confident
- ○ Confident
- ○ Somewhat confident
- ○ Not confident

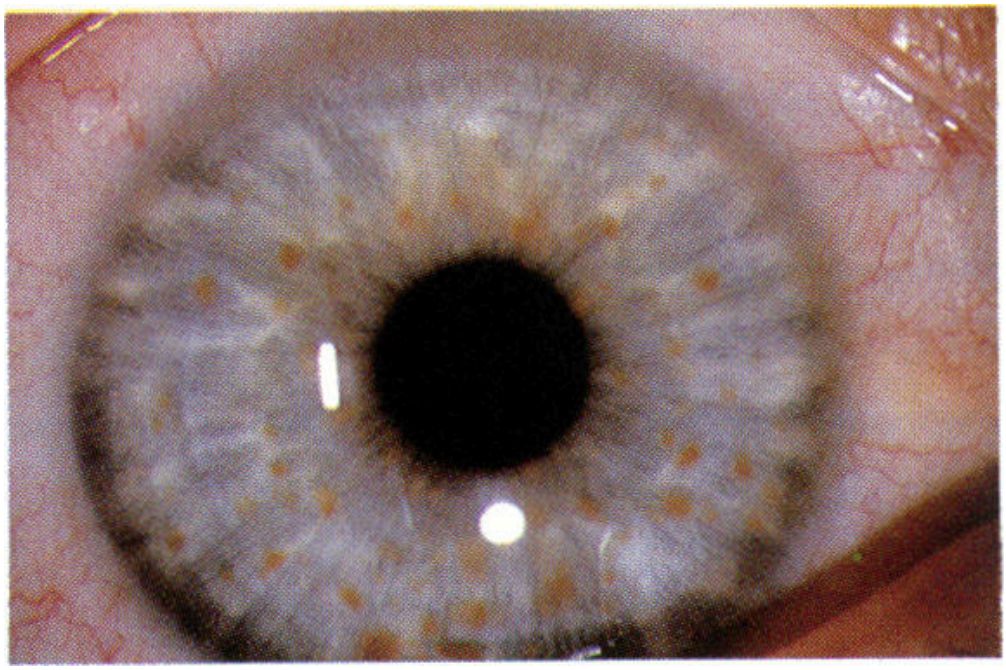


This medical image shows a patient with Kabuki syndrome. Please evaluate whether the following clinical feature is visible and accurately describes what you observe in the image -- Feature: sparse lateral eyebrow.

- Yes
- No

How confident are you with your answer?

- Highly confident
- Confident
- Somewhat confident
- Not confident



This medical image shows a patient with Neurofibromatosis type 1. Please evaluate whether the following clinical feature is visible and accurately describes what you observe in the image -- Feature: Lisch nodules.

- ○ Yes
- ○ No

How confident are you with your answer?

- ○ Highly confident
- ○ Confident
- ○ Somewhat confident
- ○ Not confident



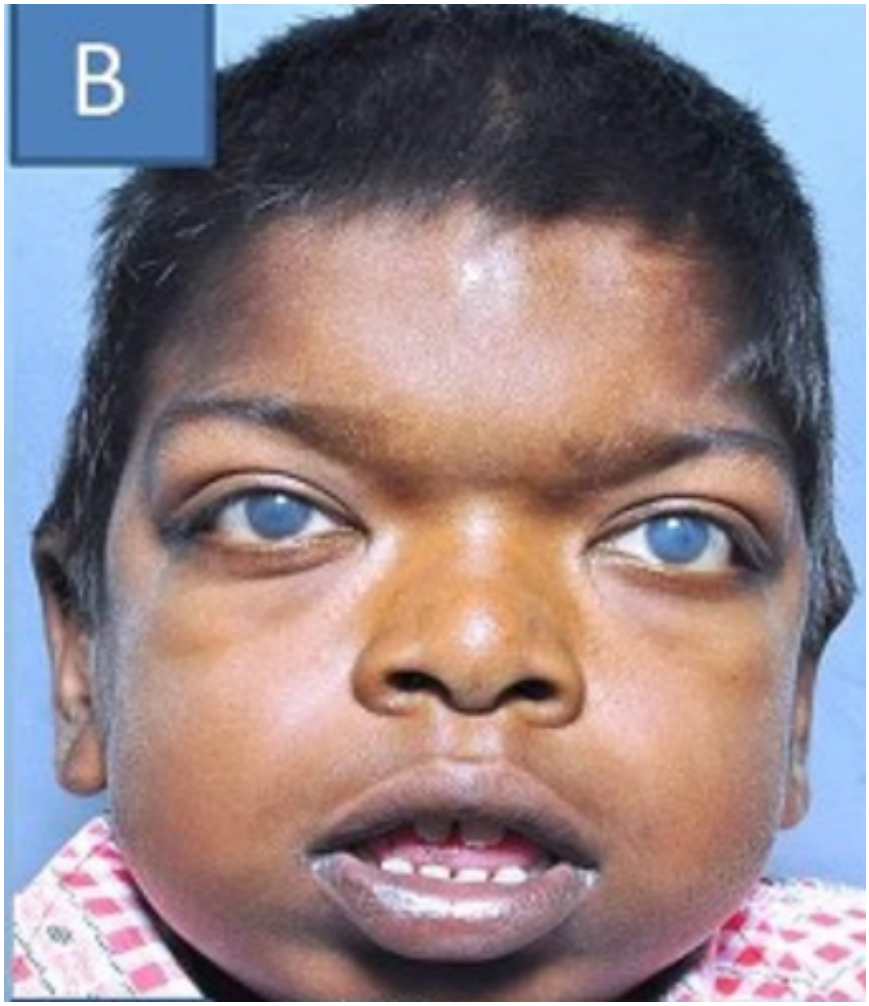
This medical image shows a patient with Alagille syndrome. Please evaluate whether the following clinical feature is visible and accurately describes what you observe in the image -- Feature: proptosis.

- ○ Yes
- ○ No

How confident are you with your answer?

- ○ Highly confident

- ○ Confident
- ○ Somewhat confident
- ○ Not confident



This medical image shows a patient with Hurler syndrome. Please evaluate whether the following clinical feature is visible and accurately describes what you observe in the image -- Feature: prematurely aged appearance.

- ○ Yes
- ○ No

How confident are you with your answer?

- ○ Highly confident
- ○ Confident
- ○ Somewhat confident
- ○ Not confident

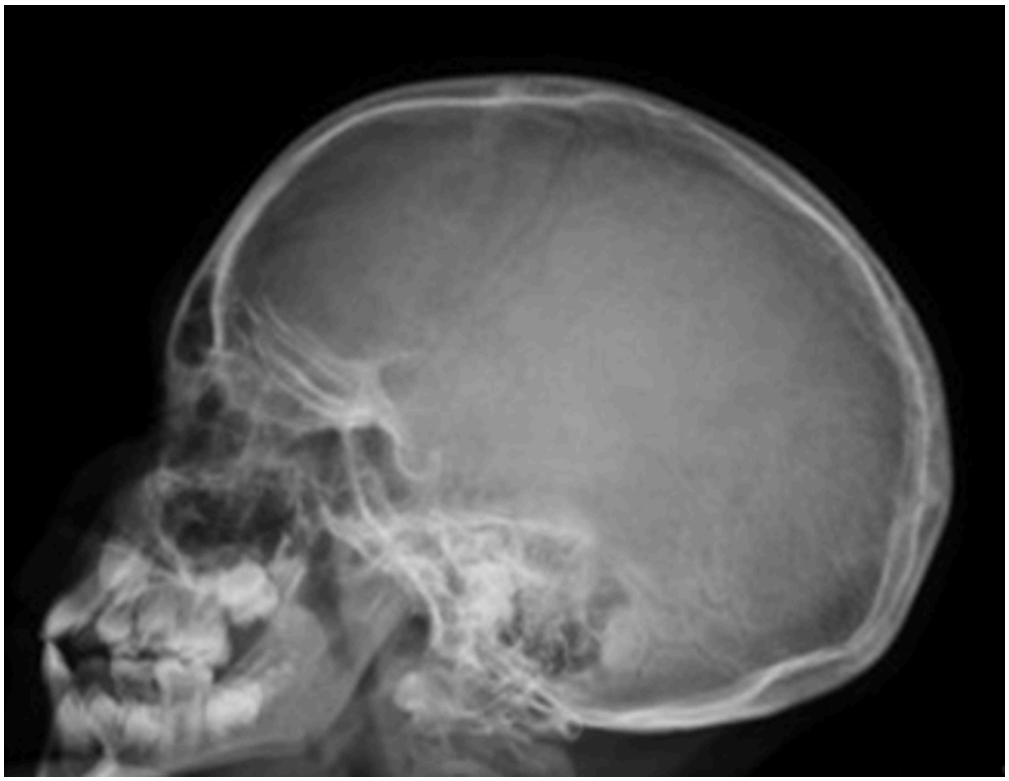


This medical image shows a patient with Neurofibromatosis type 1. Please evaluate whether the following clinical feature is visible and accurately describes what you observe in the image -- Feature: neurofibromas.

- Yes
- No

How confident are you with your answer?

- Highly confident
- Confident
- Somewhat confident
- Not confident



This medical image shows a patient with Osteogenesis imperfecta, type XI. Please evaluate

whether the following clinical feature is visible and accurately describes what you observe in the image -- Feature: Wormian bones.

- Yes
- No

How confident are you with your answer?

- Highly confident
- Confident
- Somewhat confident
- Not confident



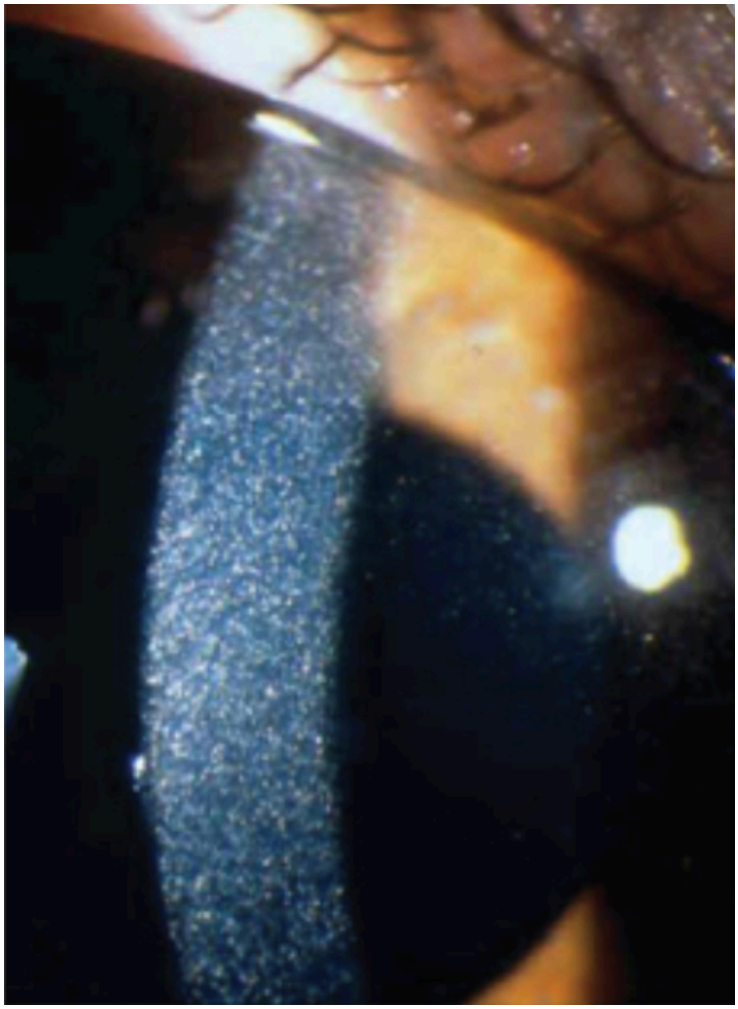
This medical image shows a patient with Primary microcephaly. Please evaluate whether the following clinical feature is visible and accurately describes what you observe in the image -- Feature: prominent metopic ridge.

- ○ Yes
- ○ No

How confident are you with your answer?

- ○ Highly confident
- ○ Confident

- ○ Somewhat confident
- ○ Not confident



This medical image shows a patient with Cystinosis. Please evaluate whether the following clinical feature is visible and accurately describes what you observe in the image -- Feature: cataract.

- ○ Yes
- ○ No

How confident are you with your answer?

- Highly confident
- Confident
- Somewhat confident
- Not confident



This medical image shows a patient with Ataxia telangiectasia. Please evaluate whether the

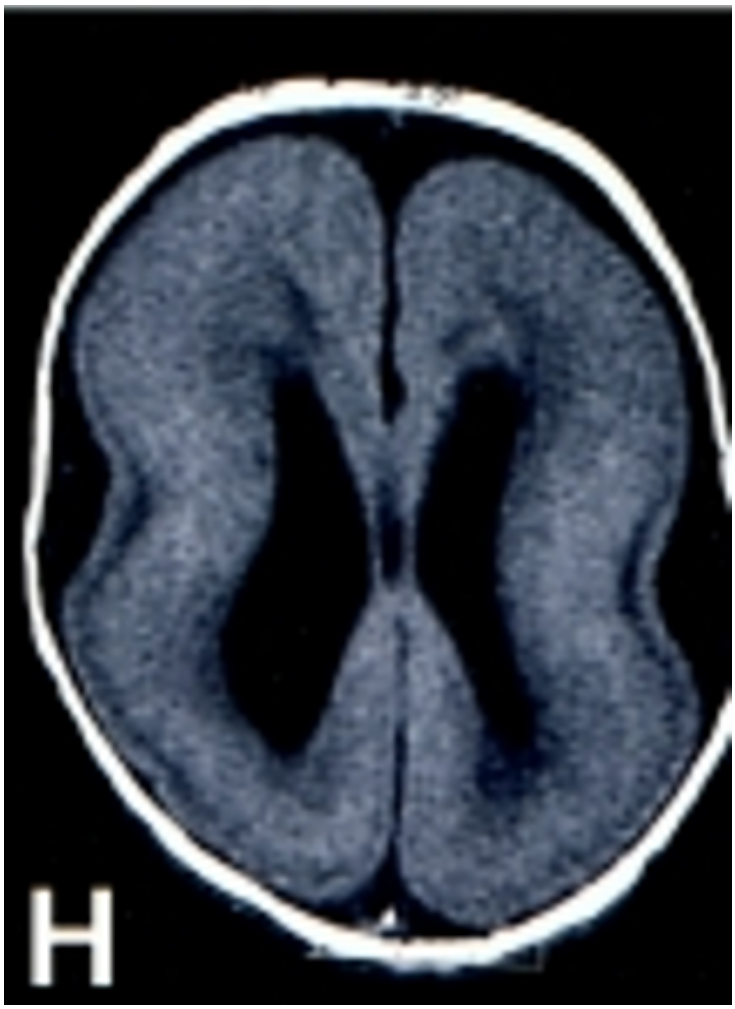
following clinical feature is visible and accurately describes what you observe in the image --

Feature: ocular telangiectasia

- Yes
- No

How confident are you with your answer?

- Highly confident
- Confident
- Somewhat confident
- Not confident



This medical image shows a patient with Miller-Dieker syndrome. Please evaluate whether the following clinical feature is visible and accurately describes what you observe in the image -- Feature: lissencephaly.

- ○ Yes
- ○ No

How confident are you with your answer?

- ○ Highly confident

- ○ Confident
- ○ Somewhat confident
- ○ Not confident



This medical image shows a patient with Huntington disease. Please evaluate whether the following clinical feature is visible and accurately describes what you observe in the image -- Feature: cortical tubers.

- ○ Yes
- ○ No

How confident are you with your answer?

- ○ Highly confident
- ○ Confident
- ○ Somewhat confident
- ○ Not confident



This medical image shows a patient with Van der Woude syndrome. Please evaluate whether the following clinical feature is visible and accurately describes what you observe in the image -- Feature: lip freckle.

- ○ Yes

- ○ No

How confident are you with your answer?

- ○ Highly confident
- ○ Confident
- ○ Somewhat confident
- ○ Not confident



This medical image shows a patient with Hallermann-Streiff syndrome. Please evaluate whether the following clinical feature is visible and accurately describes what you observe in the image -- Feature: narrow mouth.

- ○ Yes
- ○ No

How confident are you with your answer?

- ○ Highly confident
- ○ Confident
- ○ Somewhat confident
- ○ Not confident



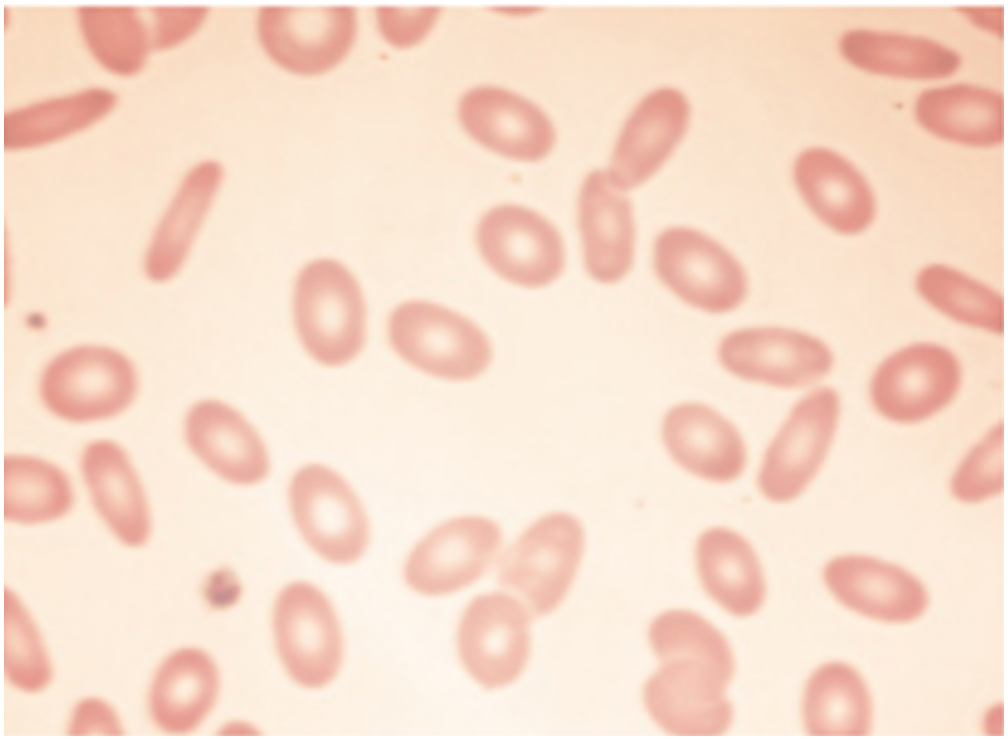
This medical image shows a patient with Waardenburg syndrome. Please evaluate whether the following clinical feature is visible and accurately describes what you observe in the image -- Feature: heterochromia iridis.

- ○ Yes
- ○ No

How confident are you with your answer?

- ○ Highly confident

- ○ Confident
- ○ Somewhat confident
- ○ Not confident



This medical image shows a patient with Abetalipoproteinemia. Please evaluate whether the following clinical feature is visible and accurately describes what you observe in the image -- Feature: elliptocytosis.

- ○ Yes
- ○ No

How confident are you with your answer?

- ○ Highly confident
- ○ Confident
- ○ Somewhat confident
- ○ Not confident



This medical image shows a patient with Gaucher disease. Please evaluate whether the following

clinical feature is visible and accurately describes what you observe in the image -- Feature: femoral bowing.

- ○ Yes
- ○ No

How confident are you with your answer?

- ○ Highly confident
- ○ Confident
- ○ Somewhat confident
- ○ Not confident



This medical image shows a patient with Ectrodactyly, ectodermal dysplasia, and cleft lip/palate syndrome. Please evaluate whether the following clinical feature is visible and accurately describes what you observe in the image -- Feature: ectrodactyly.

- ○ Yes
- ○ No

How confident are you with your answer?

- ○ Highly confident
- ○ Confident
- ○ Somewhat confident
- ○ Not confident

Set 2A



This medical image shows a patient with Mulibrey nanism syndrome. Please evaluate whether the following clinical feature is visible and accurately describes what you observe in the image -- Feature: J-shaped sella turcica.

- ○ Yes
- ○ No

How confident are you with your answer?

- ○ Highly confident
- ○ Confident
- ○ Somewhat confident
- ○ Not confident



This medical image shows a patient with Seckel syndrome. Please evaluate whether the following clinical feature is visible and accurately describes what you observe in the image -- Feature: flat face.

- ○ Yes
- ○ No

How confident are you with your answer?

- ○ Highly confident
- ○ Confident
- ○ Somewhat confident
- ○ Not confident



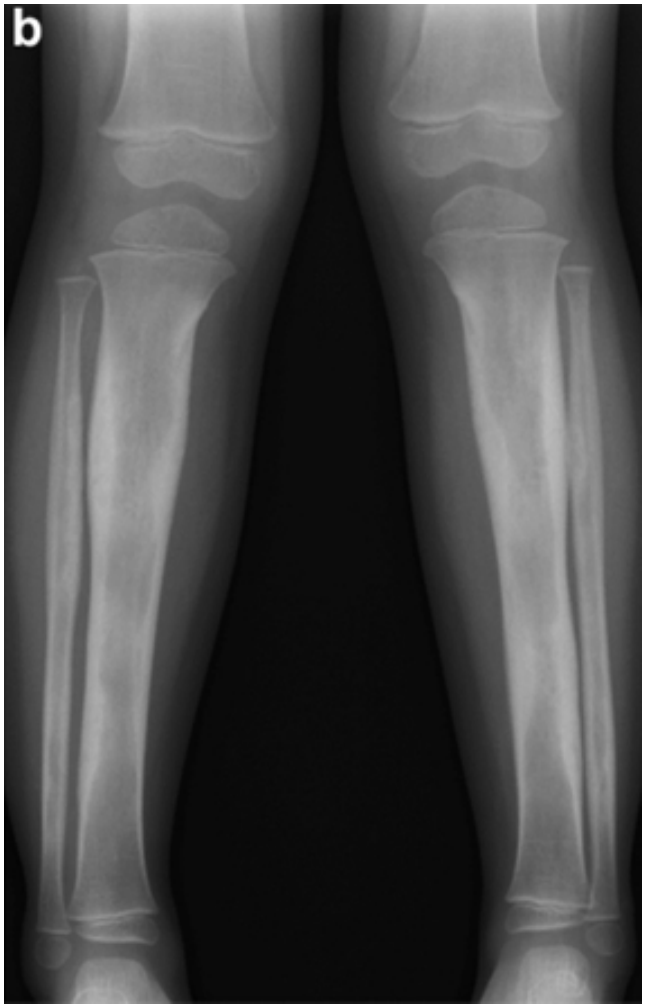
This medical image shows a patient with Autosomal recessive polycystic kidney disease. Please evaluate whether the following clinical feature is visible and accurately describes what you observe in the image -- Feature: Nephroblastoma.

- ○ Yes
- ○ No

How confident are you with your answer?

- ○ Highly confident

- ○ Confident
- ○ Somewhat confident
- ○ Not confident

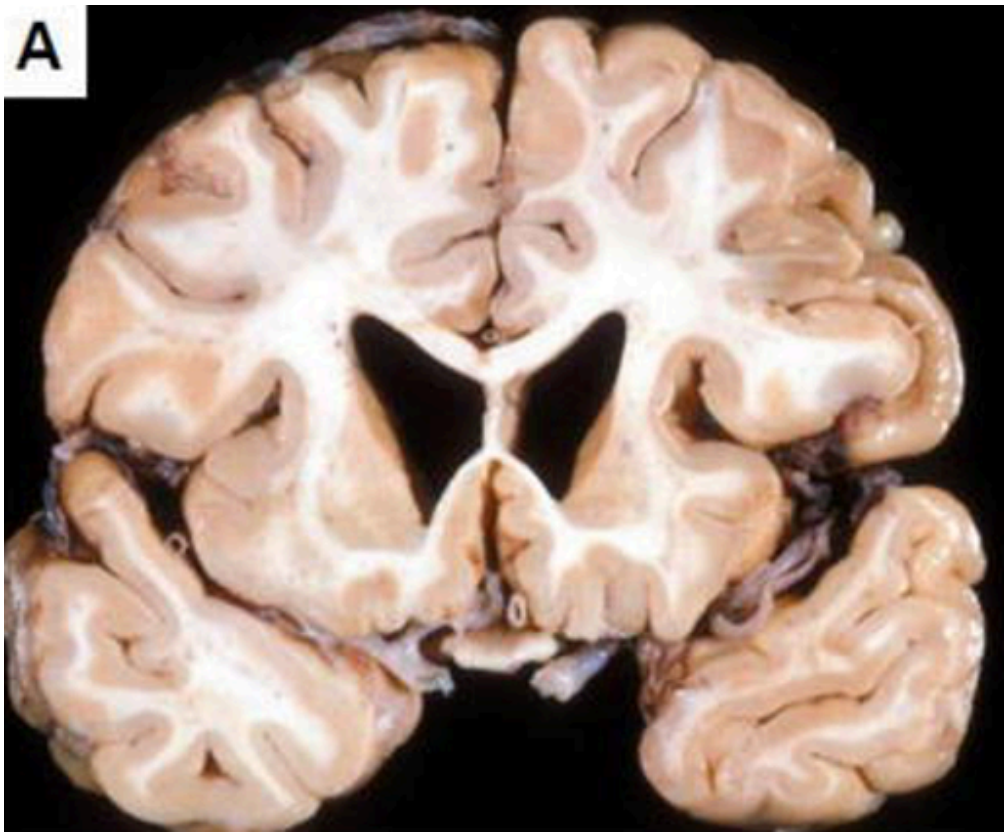


This medical image shows a patient with Cartilage-hair hypoplasia. Please evaluate whether the following clinical feature is visible and accurately describes what you observe in the image -- Feature: tibial bowing.

- ○ Yes
- ○ No

How confident are you with your answer?

- ○ Highly confident
- ○ Confident
- ○ Somewhat confident
- ○ Not confident



This medical image shows a patient with Huntington disease. Please evaluate whether the

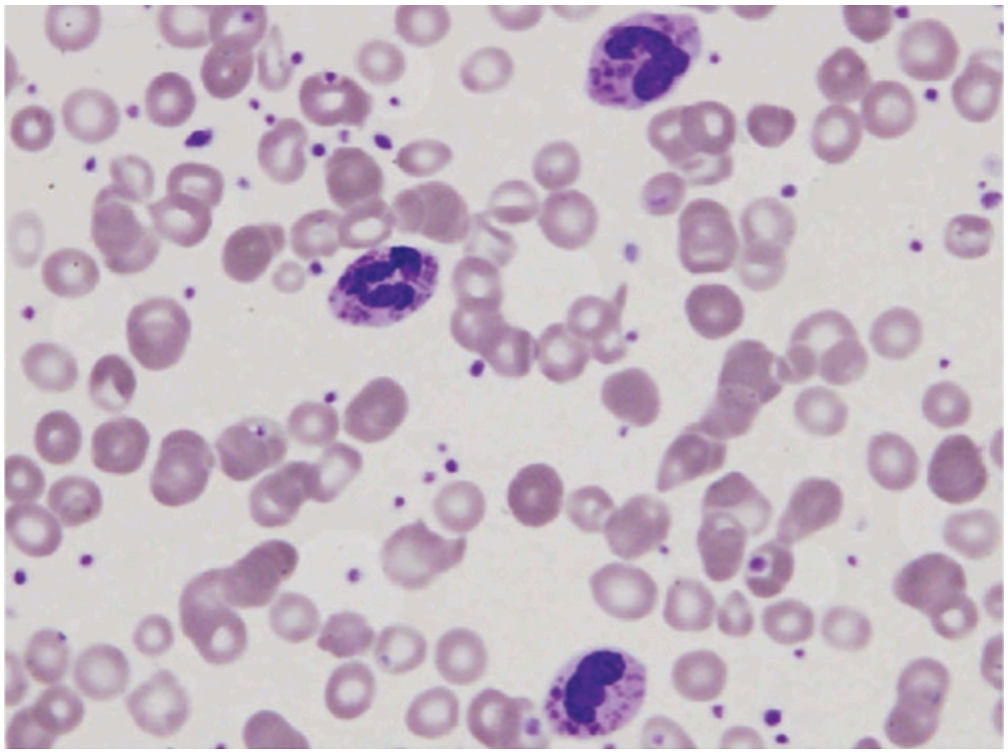
following clinical feature is visible and accurately describes what you observe in the image --

Feature: caudate atrophy.

- Yes
- No

How confident are you with your answer?

- Highly confident
- Confident
- Somewhat confident
- Not confident



This medical image shows a patient with Wiskott-Aldrich syndrome. Please evaluate whether the following clinical feature is visible and accurately describes what you observe in the image -- Feature: giant neutrophil granules.

- ○ Yes
- ○ No

How confident are you with your answer?

- ○ Highly confident
- ○ Confident
- ○ Somewhat confident
- ○ Not confident



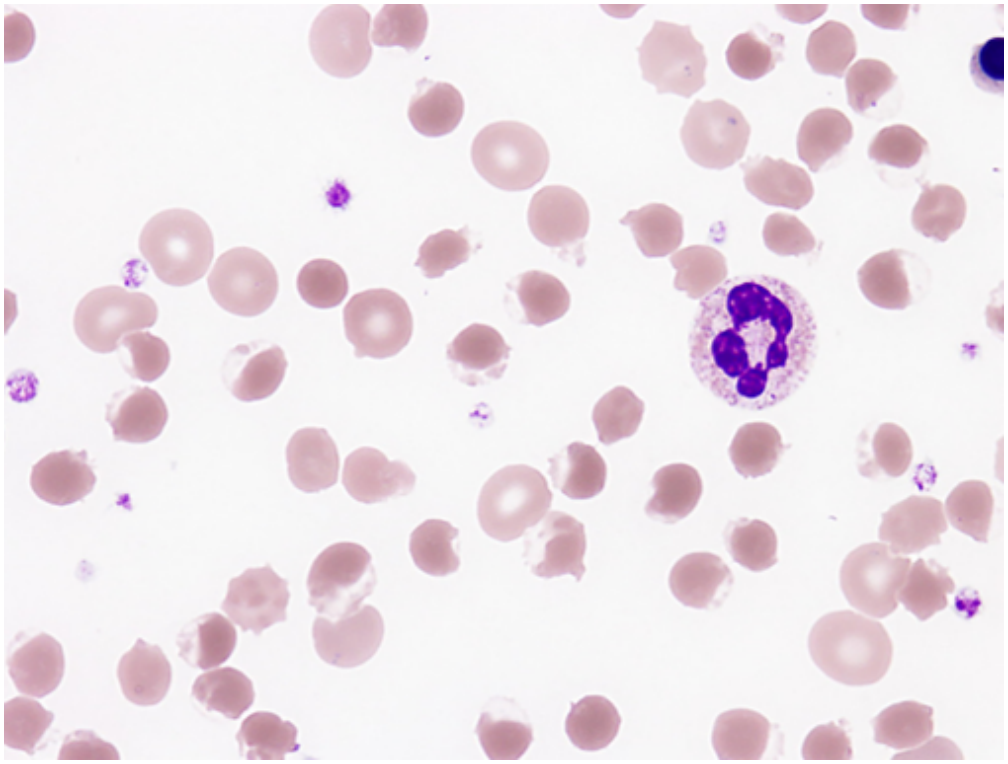
This medical image shows a patient with Bloom syndrome. Please evaluate whether the following clinical feature is visible and accurately describes what you observe in the image -- Feature: chromosomal breakage induced by crosslinking agents.

- ○ Yes
- ○ No

How confident are you with your answer?

- ○ Highly confident

- ○ Confident
- ○ Somewhat confident
- ○ Not confident



This medical image shows a patient with Erythrocytosis. Please evaluate whether the following clinical feature is visible and accurately describes what you observe in the image -- Feature: polycythemia.

- ○ Yes
- ○ No

How confident are you with your answer?

- ○ Highly confident
- ○ Confident
- ○ Somewhat confident
- ○ Not confident



This medical image shows a patient with Thrombocytopenia-absent radius syndrome. Please evaluate whether the following clinical feature is visible and accurately describes what you observe in the image -- Feature: absent radius.

- ○ Yes
- ○ No

How confident are you with your answer?

- ○ Highly confident
- ○ Confident
- ○ Somewhat confident
- ○ Not confident



This medical image shows a patient with Jeune syndrome (Short-rib thoracic dysplasia). Please evaluate whether the following clinical feature is visible and accurately describes what you observe in the image -- Feature: situs inversus totalis.

- ○ Yes
- ○ No

How confident are you with your answer?

- ○ Highly confident
- ○ Confident
- ○ Somewhat confident
- ○ Not confident



This medical image shows a patient with Craniofrontonasal dysplasia. Please evaluate whether the following clinical feature is visible and accurately describes what you observe in the image -- Feature: low hanging columella.

- Yes
- No

How confident are you with your answer?

- Highly confident

- ○ Confident
- ○ Somewhat confident
- ○ Not confident



This medical image shows a patient with Orofaciodigital syndrome. Please evaluate whether the following clinical feature is visible and accurately describes what you observe in the image -- Feature: polydactyly.

- ○ Yes
- ○ No

How confident are you with your answer?

- ○ Highly confident
- ○ Confident
- ○ Somewhat confident
- ○ Not confident



This medical image shows a patient with Adams-Oliver syndrome. Please evaluate whether the

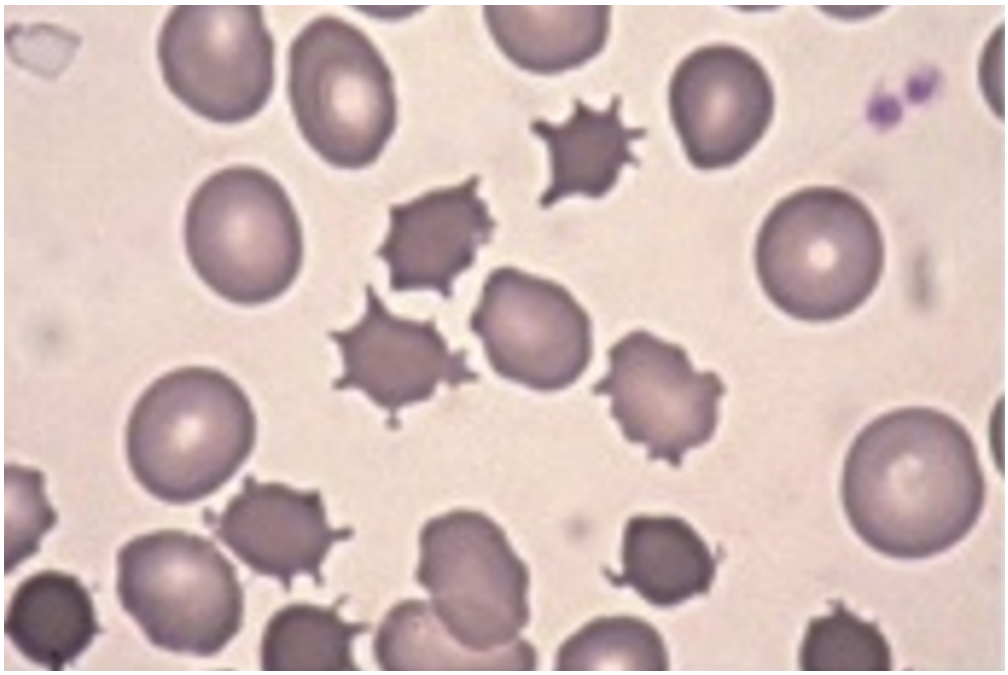
following clinical feature is visible and accurately describes what you observe in the image --

Feature: aplasia cutis congenita.

- Yes
- No

How confident are you with your answer?

- Highly confident
- Confident
- Somewhat confident
- Not confident



This medical image shows a patient with Sideroblastic anemia and spinocerebellar ataxia. Please

evaluate whether the following clinical feature is visible and accurately describes what you observe in the image -- Feature: acanthocytosis.

- Yes
- No

How confident are you with your answer?

- Highly confident
- Confident
- Somewhat confident
- Not confident



This medical image shows a patient with Rubinstein-Taybi syndrome. Please evaluate whether the following clinical feature is visible and accurately describes what you observe in the image -- Feature: triphalangeal thumb.

- ○ Yes
- ○ No

How confident are you with your answer?

- ○ Highly confident
- ○ Confident

- ○ Somewhat confident
- ○ Not confident



This medical image shows a patient with Townes-Brocks syndrome. Please evaluate whether the following clinical feature is visible and accurately describes what you observe in the image -- Feature: preauricular skin tag.

- ○ Yes
- ○ No

How confident are you with your answer?

- Highly confident
- Confident
- Somewhat confident
- Not confident



This medical image shows a patient with Hermansky-Pudlak Syndrome. Please evaluate whether

the following clinical feature is visible and accurately describes what you observe in the image --

Feature: ecchymosis.

- ○ Yes
- ○ No

How confident are you with your answer?

- ○ Highly confident
- ○ Confident
- ○ Somewhat confident
- ○ Not confident



This medical image shows a patient with Hermansky-Pudlak Syndrome. Please evaluate whether the following clinical feature is visible and accurately describes what you observe in the image -- Feature: xanthomatosis.

- ○ Yes
- ○ No

How confident are you with your answer?

- ○ Highly confident
- ○ Confident

- ○ Somewhat confident
- ○ Not confident

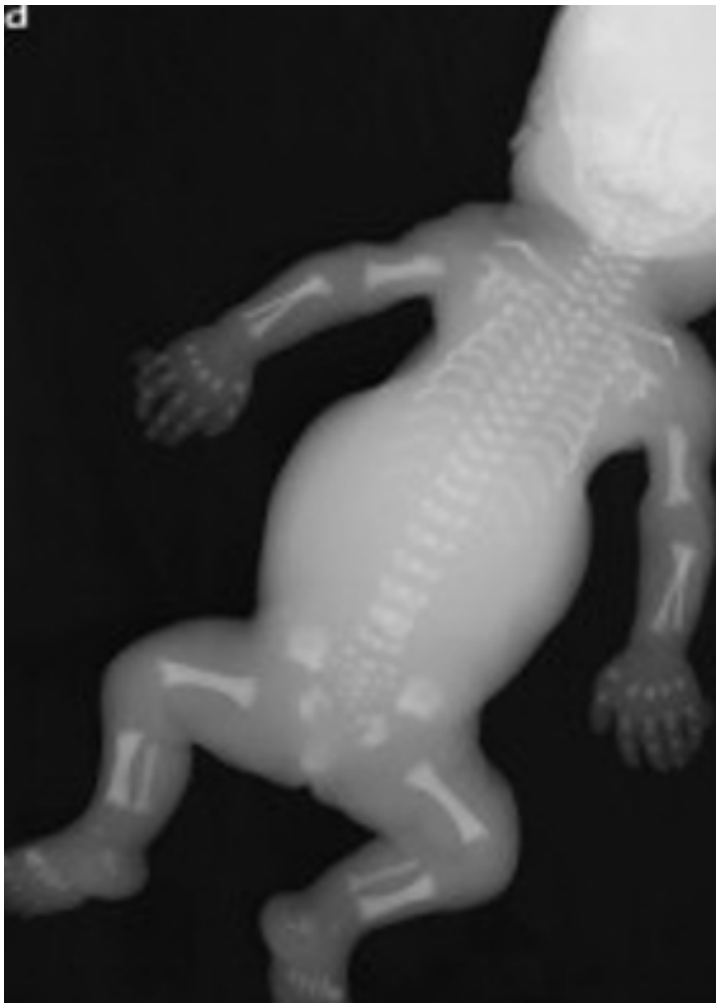


This medical image shows a patient with Peutz-Jeghers syndrome. Please evaluate whether the following clinical feature is visible and accurately describes what you observe in the image -- Feature: cleft lip.

- ○ Yes
- ○ No

How confident are you with your answer?

- ○ Highly confident
- ○ Confident
- ○ Somewhat confident
- ○ Not confident



This medical image shows a patient with Xia-Gibbs syndrome. Please evaluate whether the following clinical feature is visible and accurately describes what you observe in the image -- Feature: enlarged thorax.

- Yes
- No

How confident are you with your answer?

- Highly confident
- Confident

- ○ Somewhat confident
- ○ Not confident

Follow-up Questions

How often did you use your knowledge of the stated genetic condition to identify whether the clinical feature was present in the image?

- ○ I never considered the stated genetic condition.
- ○ I sometimes considered the stated genetic condition.
- ○ I mostly considered the stated genetic condition.
- ○ I always considered the stated genetic condition.

Were the genetic conditions correctly depicted in the images?

- ○ All were correct.
- ○ Most were correct.
- ○ A few appeared incorrect.
- ○ Several appeared incorrect.
- ○ All were incorrect.

Considering all questions collectively, were the stated clinical features typical findings of the stated

genetic condition?

- All clinical features were typical findings of the stated genetic conditions.
- Most clinical features were typical findings of the stated genetic condition, a few were not typical.
- About half of the clinical features were typical findings of the stated genetic conditions.
- Most were not typical clinical features of the stated genetic conditions, a few were typical.
- All were not typical clinical features of the stated genetic conditions.

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